

Insurance Complaint & Premium Analysis Report

1. Objective

The objective of this analysis is to:

1. Evaluate **complaint patterns** across insurance companies
 2. Analyze **premium trends over time**
 3. Compare performance across **5 selected companies**
 4. Provide **actionable insights and recommendations**
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2. Dataset Understanding

The dataset contains insurance company performance metrics provided by DFS (New York State), including:

Key Fields

- **NAIC:** Unique company identifier
 - **Company_Name:** Insurance provider
 - **Ratio:** Complaint ratio (Upheld / Premiums)
 - **Upheld Complaints:** Valid complaints
 - **Question of Fact Complaints (QoF):** Disputed cases
 - **Not Upheld Complaints:** Rejected/withdrawn complaints
 - **Total Complaints:** Sum of all complaint types
 - **Premiums_Written:** Revenue proxy (in millions)
 - **Ranking:** Performance rank
 - **Filing_Year:** Year of complaint closure
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3. Data Preparation Steps

✓ Step 1: Data Cleaning

- Removed **FIELD11 & FIELD12** (contained only NULL values)
 - Verified:
 - No missing values in key columns
 - Correct data types (numeric, text)
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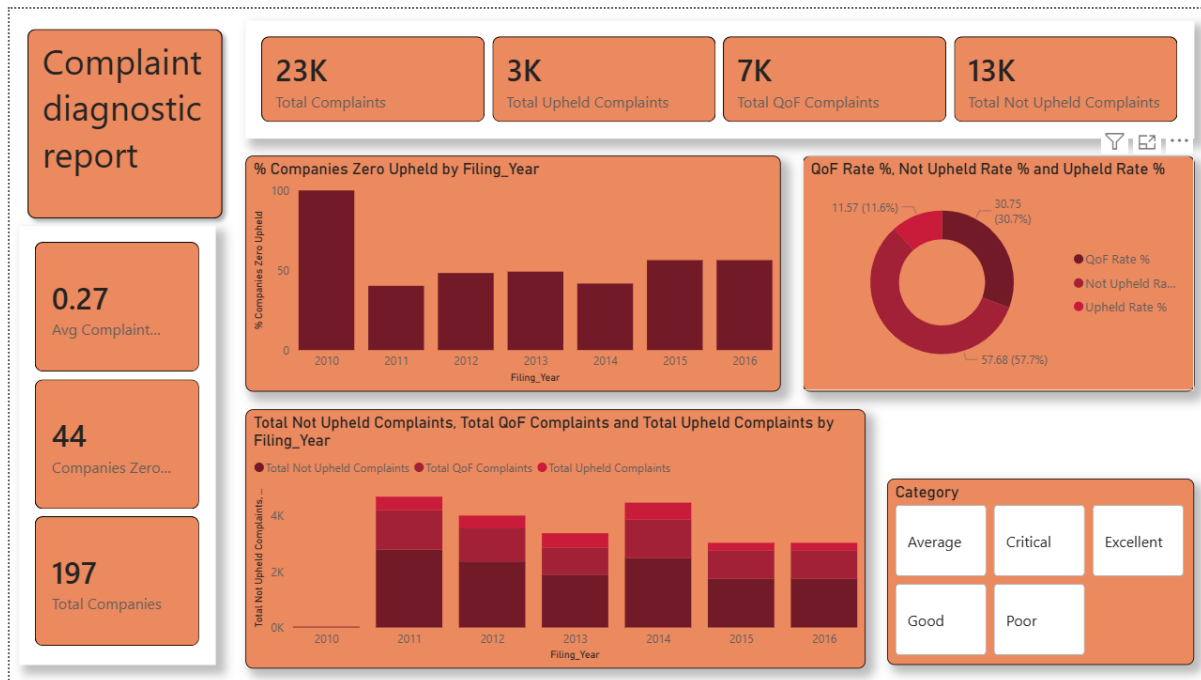
✓ Step 2: Data Transformation

- Created calculated fields:
 - Complaint categories (Excellent, Good, Average, Poor, Critical)
 - Aggregated measures (Total Premiums, Avg Premium, % Companies Zero Upheld, Avg Complaint Ratio etc.)

✓ Step 3: Data Modeling

- Used **Filing_Year** as time dimension
- Created relationships where required
- Built measures using DAX

5. Dashboard 1: Complaint Diagnostic Report



Purpose

Understand complaint behavior and company performance

Key Insights

1. Complaint Distribution

- Majority complaints fall under:
 - **Not Upheld (~57%)**
 - **QoF (~30%)**

- **Upheld (~11%)**

Interpretation:

- Most complaints are either invalid or inconclusive
 - Only a small portion are genuinely valid
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2. Trend Over Years

- Complaint volume shows:
 - Peak around **2011–2014**
 - Gradual stabilization afterward
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3. Zero Upheld Companies

- A significant number of companies have:
 - **0 upheld complaints**

Indicates:

- Strong compliance OR
 - Low claim scrutiny
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Recommendation

Companies with:

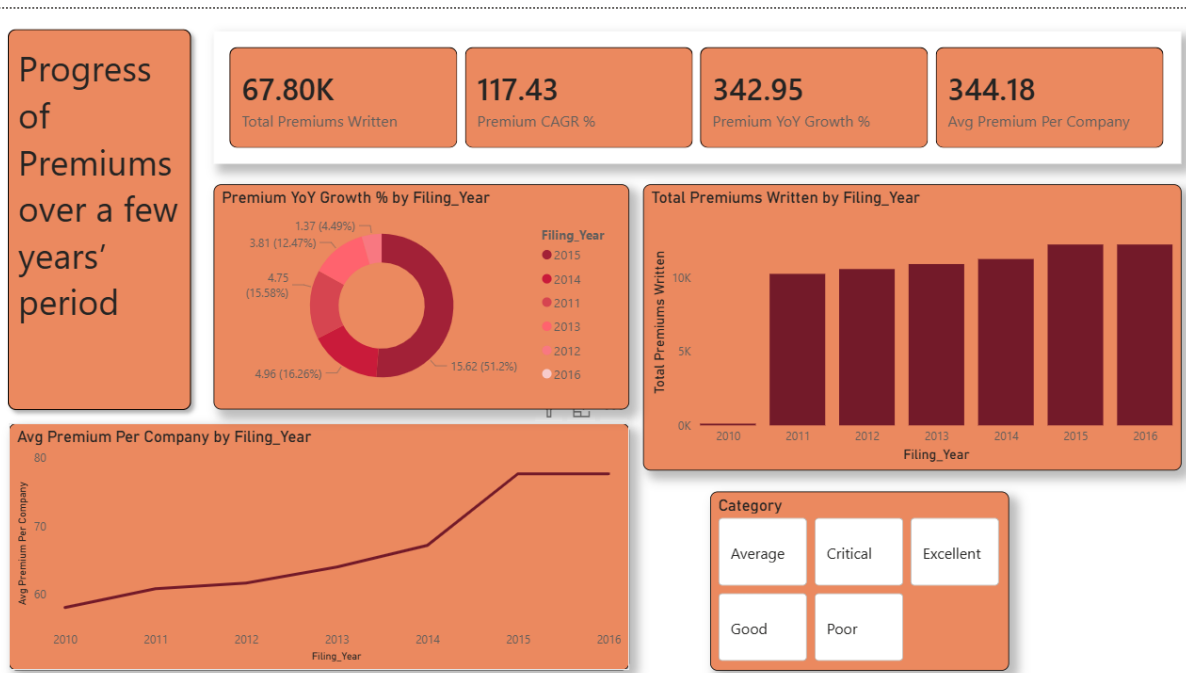
- **Low complaint ratio**
- **Zero upheld complaints**

can be considered **high-performing**

Example:

- *Bankers Standard Insurance Company*
 - *Great Northern Insurance Company*
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6. Dashboard 2: Premium Progress Analysis



Purpose

Analyze revenue growth and identify drivers

Key Insights

1. Premium Growth Trend

- Steady increase from **2010 → 2016**
- Significant rise after **2014**

Indicates:

- Market expansion
- Increased policy sales

2. YoY Growth

- Growth is **positive but fluctuating**
- Peak growth observed in later years

3. CAGR (~117%)

- Indicates strong long-term growth

Note:

- High CAGR may be due to:

- Low base year value
- Rapid scaling phase

Possible Causes of Growth

Business Factors

- Increased customer base
- Competitive pricing

Market Factors

- Economic growth
- Higher vehicle ownership

Regulatory Factors

- Policy changes in NYS insurance laws

Future Trend Prediction

Based on historical data:

Premiums are likely to:

- Continue **moderate upward growth**
- Stabilize as market matures

Observed Pattern (from your dashboard)

- Premiums show a **steady upward trend (2010–2016)**
- Noticeable **acceleration after 2014**
- YoY growth is **positive but uneven**

This indicates:

- Growth is **not purely linear**
- External and operational factors are influencing premiums

Potential Drivers of Premium Fluctuations

1. Claims Frequency & Complaint Behavior

Observation:

- Years with higher complaints (especially QoF & Not Upheld) show variation in premium growth

Interpretation:

- Increased complaints may indicate:
 - Higher claim activity
 - Customer dissatisfaction
- Insurers may respond by:
 - Adjusting pricing
 - Tightening underwriting policies

Impact:

Higher claims → Higher risk → Increase in premiums

2. Regulatory Changes (NYS Context)

Observation:

- Dataset is regulated by DFS (New York State)

Possible influence:

- Changes in:
 - Claim settlement rules
 - No-fault insurance policies
 - Consumer protection laws

Impact:

Stricter regulations → Higher compliance cost → Increased premiums

3. Economic Conditions

Factors:

- Inflation
- Income levels
- Vehicle ownership trends

Interpretation:

- Strong economy → More vehicles → More policies → Higher premiums
- Weak economy → Reduced demand

Impact:

Economic growth → Increase in premiums written

4. Market Dynamics (Insurance Industry)

Key drivers:

- Competition among insurers
- Entry of new players
- Pricing strategies

Observation:

- Companies with higher premiums dominate the market

Impact:

Competitive pricing → Short-term fluctuation in premiums

Market expansion → Long-term growth

5. Company-Specific Strategy

Different companies show different trends:

- Some companies:
 - Aggressively increase premiums (growth strategy)
- Others:
 - Maintain stability (risk-controlled approach)

Impact:

Business strategy → Direct influence on premium growth patterns

Correlation Insight (Important)

From combined analysis:

- Premium growth ≠ directly proportional to complaints
 - However:
 - High complaint ratios may affect **customer trust**
 - Low complaint companies tend to show **stable growth**
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11. Future Premium Trend Insights

Based on Historical Data

✓ Trend Projection

- Premiums are likely to:
 - Continue **upward trend**
 - Grow at a **moderate and stable rate**
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Expected Behavior

1. Stabilization Phase

- Rapid growth (post-2014) may slow down
 - Market approaching maturity
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2. Data-Driven Pricing

- Companies will rely more on:
 - Risk analytics
 - Customer segmentation
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3. Customer-Centric Adjustments

- Reduction in complaints → better retention → stable premiums
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Potential Risks

Risk Factor	Impact
Increase in claims	Premium rise
Regulatory tightening	Cost increase
Economic slowdown	Reduced growth

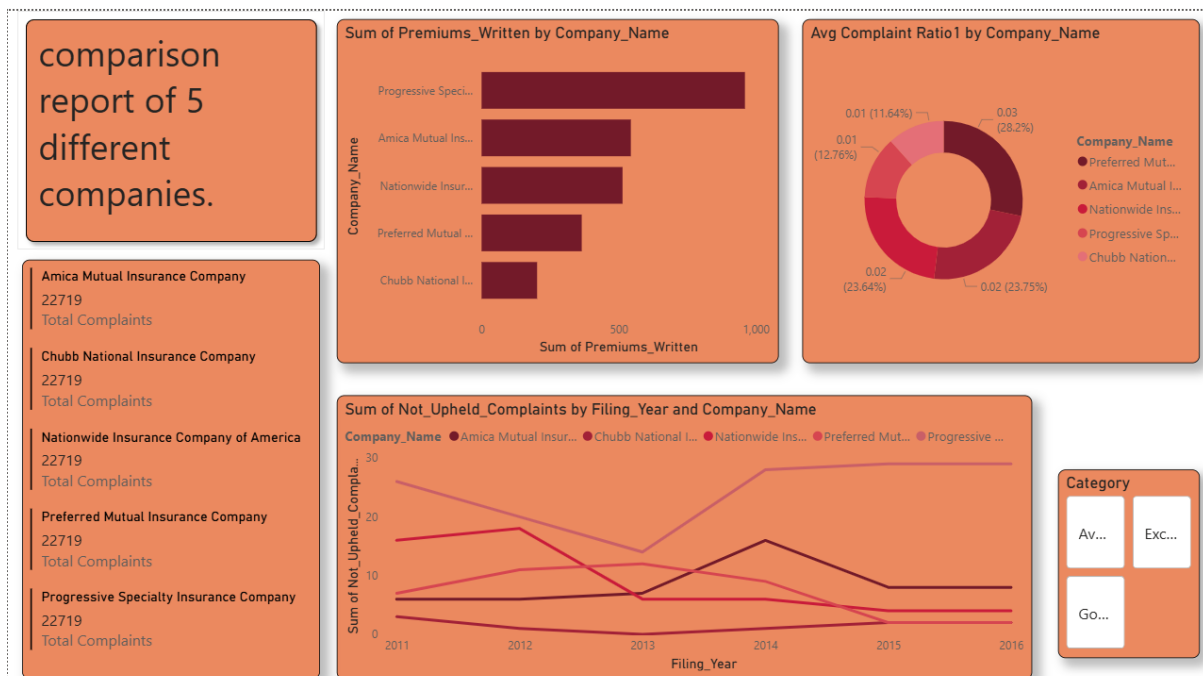
Strategic Insight

- Maintain **low complaint ratios**
 - Balance **growth with service quality**
 - Invest in **claims efficiency**
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Final Analytical Conclusion

- Premium growth is driven by a **combination of internal and external factors**
- Complaint trends act as a **risk signal**, not a direct driver
- The market shows:
 - **Strong growth potential**
 - Moving towards **stability and maturity**

7. Dashboard 3: 5-Company Comparison



Purpose

Compare top 5 companies across key metrics

Key Insights

1. Premium Contribution

- **Progressive Specialty Insurance** dominates
- Followed by:
 - Amica
 - Nationwide

2. Complaint Ratio Comparison

- Some companies show:
 - Low ratio but high volume
 - High ratio with low premium

Important distinction:

- Ratio $\neq$ volume
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3. Complaint Trends

- Certain companies show:
 - Rising complaints over time
 - Others are stable/improving
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Best Performing Company

Criteria:

- Low complaint ratio
- Stable complaint trend
- Good premium growth

Recommended:

- **Amica Mutual Insurance Company**
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Company Needing Improvement

Criteria:

- Increasing complaints
- High QoF or upheld %

Example:

- Companies with fluctuating complaint trends
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7. Key Business Insights

✓ Complaints

- Majority are **not upheld**

- Indicates strong internal validation processes
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✓ Premiums

- Strong upward trend
 - Market is expanding
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✓ Risk Indicator

- Complaint ratio is a **critical KPI**
 - Helps regulators rank companies
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8. Recommendations

For Companies

- Reduce **QoF complaints** (improve clarity in policies)
 - Focus on **customer communication**
 - Strengthen **claim processing systems**
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For Business Strategy

- Invest in **customer experience**
 - Monitor **complaint ratio alongside premiums**
 - Balance growth with service quality
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9. Conclusion

- The insurance market shows **strong growth potential**
- Complaint analysis reveals:
 - Most issues are not severe
- Top-performing companies maintain:
 - Low complaint ratios
 - Consistent service quality

Combining **premium growth + complaint control** is key to long-term success